

Continuous-field-reconstruction-from-sparse-observations-with-implicit-neural-networks-Paper-Conference

一、研究問題

如何利用非常少量、稀疏 (sparse) 的感測器資料，重建完整的連續 physical field (例如 thermal map、temperature field) ?

論文中的 field 包含:

1. Temperature
2. ocean surface temperature
3. climate field
4. velocity field

Sparse observations $\rightarrow$ Full continuous field reconstruction

二、研究動機與挑戰

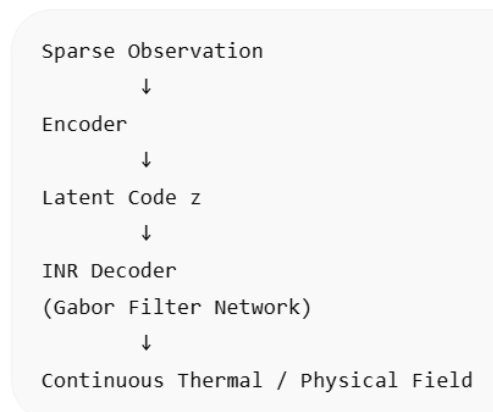
論文認為 thermal / physical field reconstruction 很困難

1. 感測器太少
2. 空間變化很複雜，傳統方法很難重建
 - (1) high-frequency variation
 - (2) nonlinear behavior
 - (3) turbulence
3. 感測器位置會變
 - (1) sensor 數量會變
 - (2) sensor 位置會變
4. 傳統 PDE/CFD 方法太難
 - (1) computationally expensive
 - (2) 難建立完整 PDE model
 - (3) 難即時運算

三、重建方法

1. MMGN (Multiplicative and Modulated Gabor Network) :屬於 Implicit Neural Representation (INR) 架構。輸入 sparse observation 後，模型學習：「連續空間函數」，而不是學習固定 pixel image。

架構：



2. Auto-decoder
3. Gabor filter
4. Multiplicative Filter Network
5. Context-aware latent code

四、輸入與輸出

1. Input: Sparse observations(sensor temperature 、sampled field value) 、Spatial coordinates (x, y) 位置資訊。
2. Output: Continuous reconstructed field(full thermal map 、continuous temperature distribution)

五、訓練資料與資料來源

1. Climate simulation data: 來自 CESM2 climate model 屬於 Simulation-based dataset
2. Satellite observation data: 來自 NASA GHRSST satellite SST dataset 屬於 Real-world observation data

六、評估指標

1. MSE (Mean Squared Error)
2. PSNR
3. SSIM
4. Relative Error

七、實驗結果

MMGN 最接近 Reference

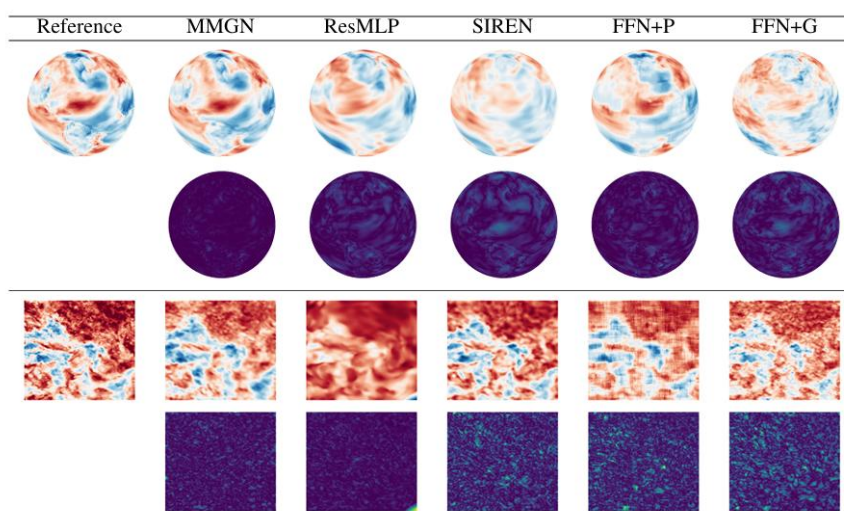


Figure 3: Visualizations of true and reconstructed fields. Global surface temperature derived from multiscale high-fidelity climate simulations and sea surface temperature assimilated using satellite imagery observations. For each dataset, the first column displays the ground truth, the first row showcases predictions from different models, and the second row presents corresponding error maps relative to the reference data. In the error maps, darker pixels indicate lower error levels.

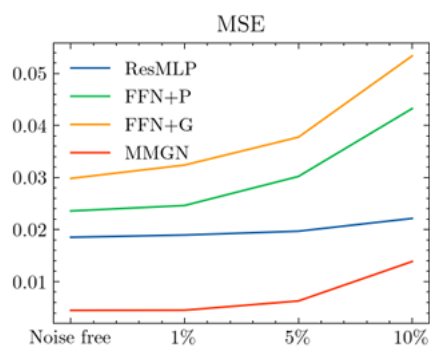


Figure 4: Model performance with different levels of noise.

八、論文限制

1. 目前只重建單一 trajectory
2. Temporal forecasting 還不完整，不是長時間 prediction。
3. 訓練成本仍高: INR 類模型，latent optimization 較慢，decoder optimization 較重。
4. 尚未真正用在 data center